Page No.
Date

03·09·2023 PHYSICS

1. Presbyopia
2. Repulsion
3. Equal to your height and Laterally inverted
4. Below the apperant depth of the fish
5. $P = \frac{1}{f} \quad \therefore f = \frac{10}{15} = +0.67$ (approx)

The prescribed lens is a convex lens

6. Given, $I = 5$ Amp $t = 30$ mins $= 1800s$

From $I = \frac{Q}{t}$ we have,

$$Q = I \times t$$
$$= (5 \times 1800)\,C$$
$$\therefore Q = 9000\,C$$

7. i) Electric cell 

ii) Electric bulb 

iii) Ammeter 

iv) Voltmeter 

v) A wire joint 

8. Given, for a convex lens - $f = 12cm$
$h_0 = 5cm$
$u = -20cm$

From $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ we have,

$$\Rightarrow \frac{1}{v} - \frac{1}{(-20)} = \frac{1}{12}$$

$$\Rightarrow \frac{1}{v} + \frac{1}{20} = \frac{1}{12}$$

$\Rightarrow \frac{1}{v} = \frac{5-3}{60}$

$\Rightarrow \frac{1}{v} = \frac{2}{60}$

$\therefore \boxed{v = 30cm}$ position of image on the other side of lens

Now,

From $\frac{h_i}{h_o} = \frac{v}{u}$ we have,

$\Rightarrow \frac{h_i}{5} = \frac{30}{-20}$

$\Rightarrow \boxed{h_i = -7.5}$

The image formed is real, Inverted and enlarged.

9. Let $V_A = 10V$ and $V_B = 5V$

Work done $= 5J$

Now,

$V_A - V_B = \frac{\text{Work done}}{Q}$

$\Rightarrow 10 - 5 = \frac{5}{Q}$

$\Rightarrow \boxed{Q = 1C}$

10. (a) Pole - It is the centre of the reflecting surface of a spherical mirror. It lies on the surface of mirror

(b) Radius of Curvature - The radius of the sphere of which the reflecting surface of spherical mirror is a part.

Represented by R

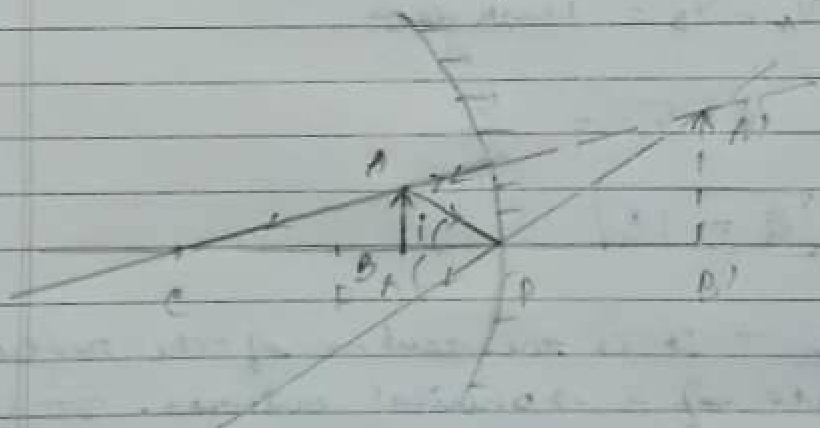
For a concave mirror it is -ve.

(c) Principal axes. - A straight line passing through the pole and the centre of curvature of a spherical mirror. The principal axis is normal to the mirror at the pole

(d) Principal focus - A point on the principal axis at which the reflected rays ~~appear~~ to meet in case of a concave mirror. Represented by F

To obtain an erect image on a concave mirror of f = 20cm we need to keep the candle b/w the P and F of the mirror i.e image distance should be less than 20 cm. but more than 0.

The image obtained will be virtual, erect and magnified.



11. In the early morning, when the sun is near
a- the horizon, the sunlight has to travel the greatest distance through the atmp. to reach us. During the long journey of sunlight, most of the blue colour and other shorter wavelength present in it

get scattered away from the line of sight. So, the light reaching us directly consists of mainly longer wavelength light i.e red light. Thus the sun appears reddish early in the morning.

b. The ability of our eye to alter the focal length of our eye lens and to adjust all the images on retina, is known as power of accomodation of the eye.

c. The least distance of distant vision of a normal eye is 25cm.